

Predictive Modeling of COVID-19 Risk in Hocus Pocus Town using Machine Learning

This presentation covers a project predicting COVID-19 risk using machine learning.

Presented by Austin Onyagada Benjamin, Data Analyst at eHealth Africa.

A by Austin Benjamin



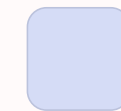


Project Overview



Goal

Predict COVID-19 likelihood from symptoms and health indicators.



Target Users

Health workers, policymakers, hospitals for early response.



Impact

Early detection enables faster isolation and reduces transmission.



Data Description

Data Source

Collected from eHealth Africa, representing Hocus Pocus town.

Dataset Size

Approximately 1,000 patient records.

Key Features

- Age, fever, cough, shortness of breath
- Comorbidities like diabetes, hypertension
- COVID-19 test outcome (positive/negative)

Exploratory Data Analysis (EDA)



60% COVID-positive

Majority of patients tested positive for COVID-19.



Symptoms prevalent

Fever and cough
strongly linked to
positive cases.



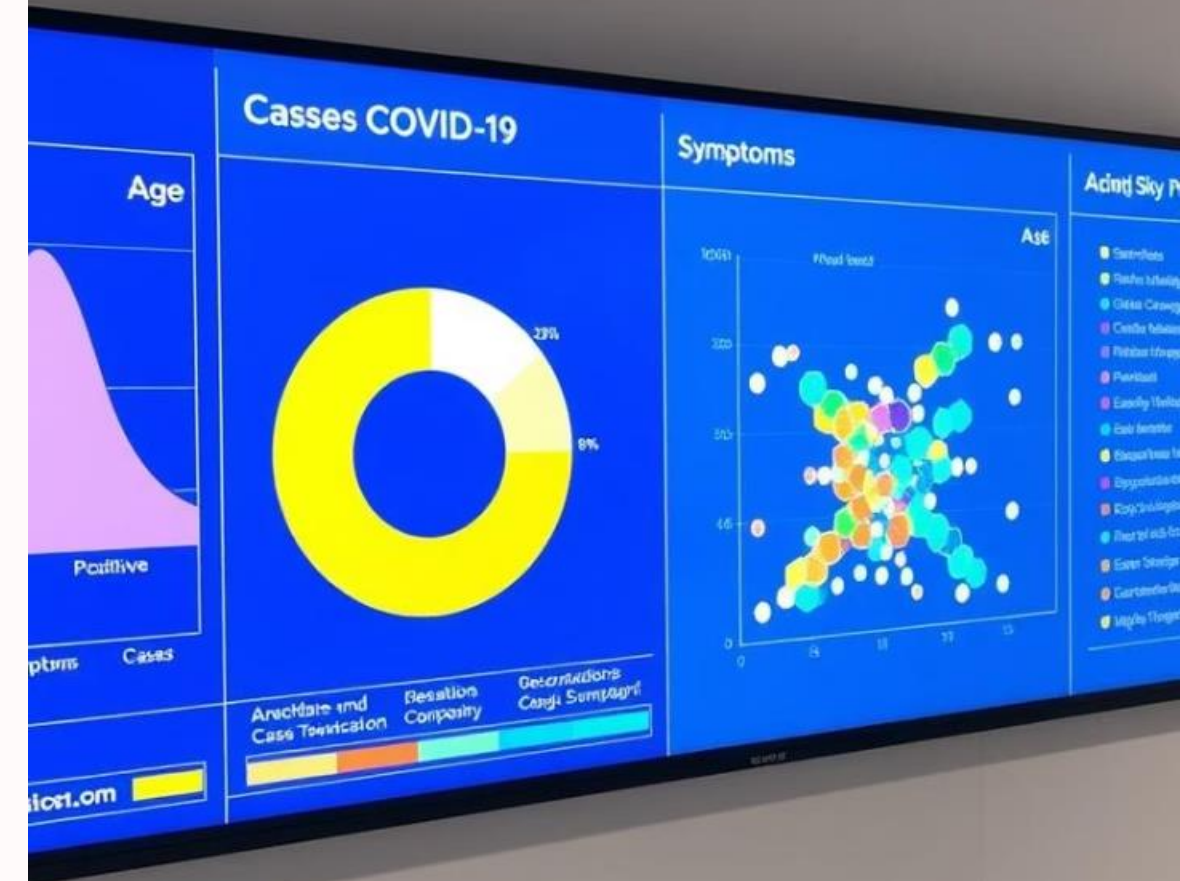
Higher risk factors

Advanced age
and
comorbidities
increased COVID
risk.



Minimal null values

Missing data was low and effectively handled.



Feature Engineering

Create Binary Flags

Converted symptom text into binary indicators for modeling.

Normalize Data

Standardized age and numerical variables for consistency.

Remove Redundancy

Eliminated highly correlated features to reduce noise.

Encode Categorical Data

Applied one-hot encoding for categorical variables.

Resulting in a clean dataset ready for machine learning models.



Machine Learning Models Used

Logistic Regression

Baseline linear model for classification.

Random Forest

Captures non-linearities and ranks feature importance.

XGBoost

Gradient boosting model achieving highest predictive performance.

All models fine-tuned with cross-validation and hyperparameter optimization.

Model Evaluation & Metrics

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Logistic Regression	0.83	0.79	0.82	0.80	0.87
Random Forest	0.89	0.86	0.88	0.87	0.93
XGBoost	0.91	0.89	0.91	0.90	0.95

XGBoost outperformed others with best balance of sensitivity and specificity.



Flask App Deployment

Web Interface

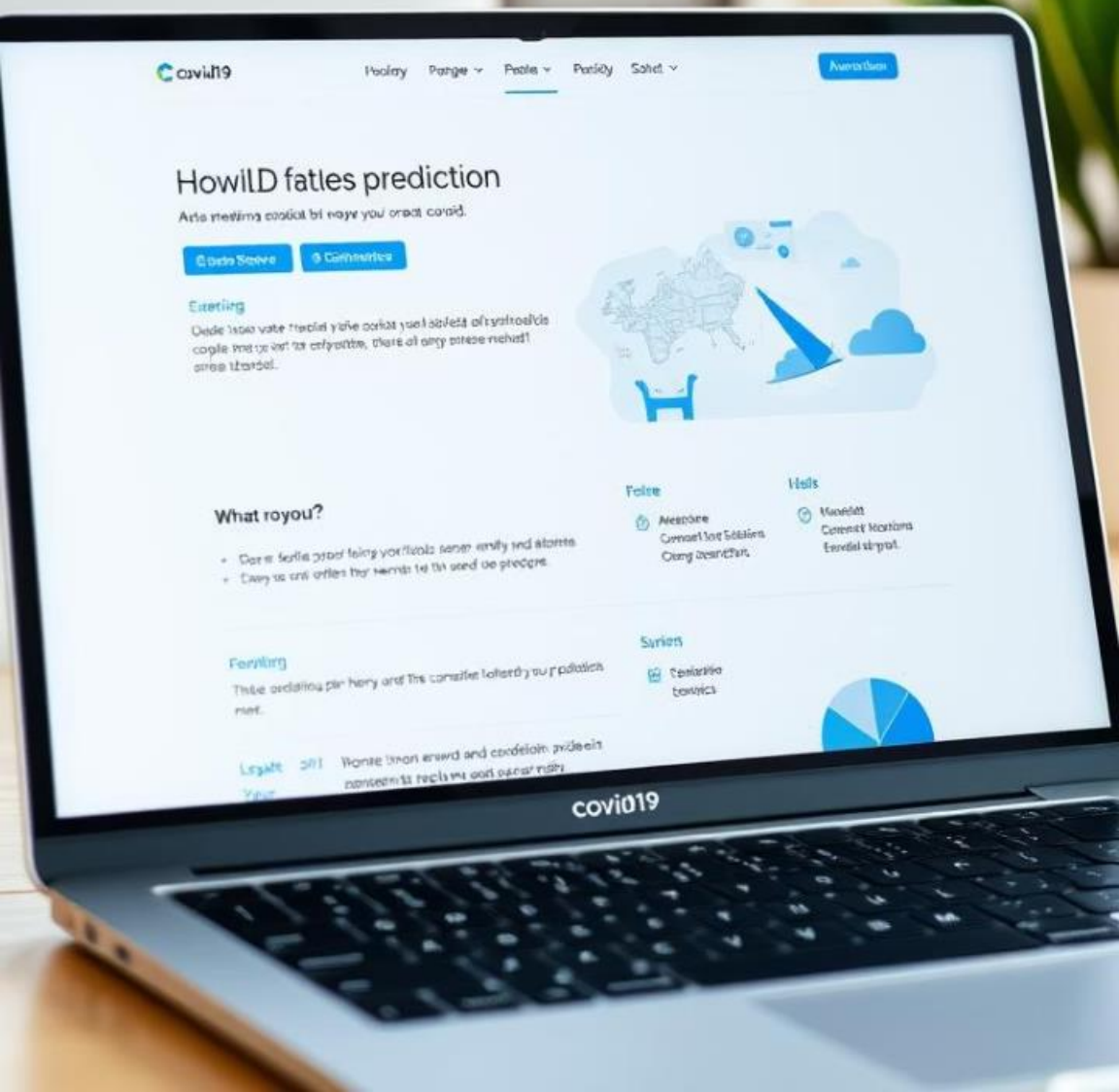
Lightweight Flask application for symptom input and model output.

Deployment

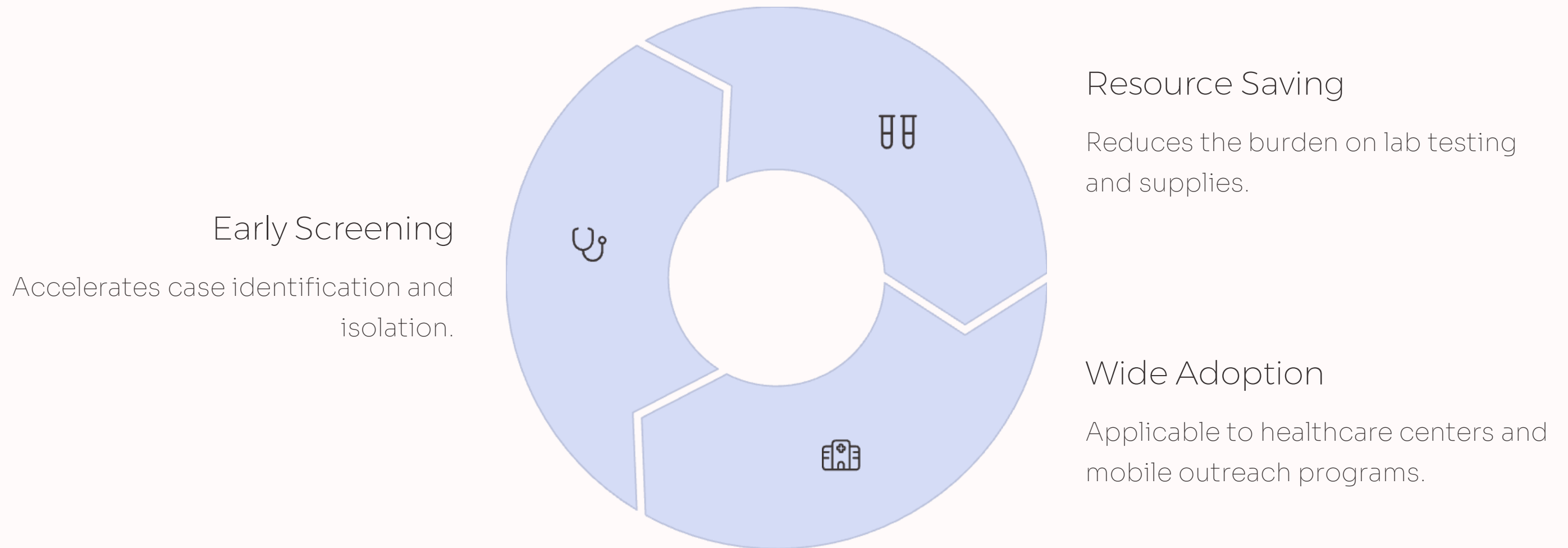
Runs locally; designed for easy cloud extension.

Interactive Use

Users get instant COVID-19 risk prediction results.



Business & Healthcare Impact



Next Steps and Conclusion

1

Enhance UI

Add multilingual support for wider accessibility.

2

Expand Dataset

Collect additional data to improve model generalization.

3

Integrate Systems

Link with health databases for real-time updates.

4

Cloud Deployment

Deploy on platforms like AWS, Render, or Streamlit.

ML delivers actionable insights, enabling data-driven healthcare decisions.

“Prevent rather than cure” – a vision powered by AI.

